

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Trigonometric Functions

Week 3 Day 2 PM



OUTLINE

- ① Trigonometric Identities
- ② Addition and Subtraction Formulas
- ③ Double-Angle Formulas
- ④ Summary

TRIGONOMETRIC IDENTITIES

FUNDAMENTAL TRIGONOMETRIC IDENTITIES

Reciprocal Identities

$$\csc x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x} \quad \cot x = \frac{1}{\tan x}$$

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1 \quad \tan^2 x + 1 = \sec^2 x \quad 1 + \cot^2 x = \csc^2 x$$

Even-Odd Identities

$$\sin(-x) = -\sin x \quad \cos(-x) = \cos x \quad \tan(-x) = -\tan x$$

Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x \quad \tan\left(\frac{\pi}{2} - x\right) = \cot x \quad \sec\left(\frac{\pi}{2} - x\right) = \csc x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x \quad \cot\left(\frac{\pi}{2} - x\right) = \tan x \quad \csc\left(\frac{\pi}{2} - x\right) = \sec x$$

SIMPLIFYING TRIGONOMETRIC EXPRESSIONS

E.g.: We can apply trigonometric identities to simplify the expression $\cos t + \tan t \sin t$ as follows:

$$\begin{aligned}\cos t + \tan t \sin t &= \cos t + \frac{\sin t}{\cos t} \sin t \\ &= \frac{\cos^2 t + \sin^2 t}{\cos t} = \frac{1}{\cos t} = \sec t.\end{aligned}$$

E.g.: We can simplify the expression $\frac{\sin x}{\cos x} + \frac{\cos x}{1 + \sin x}$ as follows:

$$\begin{aligned}\frac{\sin x}{\cos x} + \frac{\cos x}{1 + \sin x} &= \frac{\sin x(1 + \sin x) + \cos^2 x}{\cos x(1 + \sin x)} \\ &= \frac{\sin x + \sin^2 x + \cos^2 x}{\cos x(1 + \sin x)} \\ &= \frac{\sin x + 1}{\cos x(1 + \sin x)} \\ &= \frac{1}{\cos x} = \sec x.\end{aligned}$$

SIMPLIFYING TRIGONOMETRIC EXPRESSIONS

E.g.: We can simplify the expression $\cos x(\sec x - \cos x)$ as follows:

$$\begin{aligned}\cos x(\sec x - \cos x) &= \cos x \left(\frac{1}{\cos x} - \cos x \right) \\ &= \frac{\cos x}{\cos x} - \cos^2 x \\ &= 1 - \cos^2 x \\ &= \sin^2 x.\end{aligned}$$

E.g.: We can simplify the expression $\frac{1}{1-\sin x} + \frac{1}{1+\sin x}$ as follows:

$$\begin{aligned}\frac{1}{1-\sin x} + \frac{1}{1+\sin x} &= \frac{1+\sin x+1-\sin x}{(1-\sin x)(1+\sin x)} \\ &= \frac{2}{1-\sin^2 x} \\ &= \frac{2}{\cos^2 x} = 2 \sec^2 x.\end{aligned}$$

ACTIVITY 1

Simplify the following trigonometric expressions.

(a) $\tan^2 x - \sec^2 x$

(b) $\sin u + \cot u \cos u$

(c) $\frac{\cot x}{\csc x - \sin x}$

(d) $\frac{2 + \tan^2 t}{\sec^2 t} - 1$

(e) $\cot(-x) \cos(-x) + \sin(-x)$

(f) $\frac{1 + \sin t}{1 - \sin t} - \frac{1 - \sin t}{1 + \sin t}$

(g) $\sin(\cos^{-1} x)$

ACTIVITY 1: SOLUTION

(a) Since $\tan^2 x + 1 = \sec^2 x$, so $\tan^2 x - \sec^2 x = -1$.

(b)

$$\begin{aligned}\sin u + \cot u \cos u &= \sin u + \frac{\cos u}{\sin u} \cos u \\ &= \frac{\sin^2 u + \cos^2 u}{\sin u} \\ &= \frac{1}{\sin u} = \csc u.\end{aligned}$$

(c)

$$\begin{aligned}\frac{\cot x}{\csc x - \sin x} &= \frac{\frac{\cos x}{\sin x}}{\frac{1}{\sin x} - \sin x} \\ &= \frac{\cos x}{1 - \sin^2 x} \\ &= \frac{\cos x}{\cos^2 x} = \frac{1}{\cos x} = \sec x.\end{aligned}$$

ACTIVITY 1: SOLUTION

(d)

$$\begin{aligned}\frac{2 + \tan^2 t}{\sec^2 t} - 1 &= \frac{2 + \tan^2 t - \sec^2 t}{\sec^2 t} \\ &= \frac{2 + (-1)}{\sec^2 t} = \frac{1}{\sec^2 t} = \cos^2 t.\end{aligned}$$

(e)

$$\begin{aligned}\cot(-x)\cos(-x) + \sin(-x) &= (-\cot x)\cos x + (-\sin x) \\ &= -\frac{\cos x}{\sin x} \cos x - \sin x \\ &= \frac{-\cos^2 x - \sin^2 x}{\sin x} \\ &= \frac{-1}{\sin x} = -\csc x.\end{aligned}$$

ACTIVITY 1: SOLUTION

$$\begin{aligned}
 \text{(f)} \quad & \frac{1 + \sin t}{1 - \sin t} - \frac{1 - \sin t}{1 + \sin t} \\
 &= \frac{(1 + \sin t)^2 - (1 - \sin t)^2}{(1 - \sin t)(1 + \sin t)} \\
 &= \frac{(1 + \sin t + 1 - \sin t)(1 + \sin t - (1 - \sin t))}{1 - \sin^2 t} \\
 &= \frac{2(2 \sin t)}{\cos^2 t} = \frac{4}{\cos t} \frac{\sin t}{\cos t} = 4 \sec t \tan t.
 \end{aligned}$$

(g) Since the range of $\cos^{-1} x$ is $[0, \pi]$, $\sin(\cos^{-1} x) \geq 0$ and thus

$$\begin{aligned}
 \sin(\cos^{-1} x) &= \sqrt{\sin^2(\cos^{-1} x)} \\
 &= \sqrt{1 - \cos^2(\cos^{-1} x)} \\
 &= \sqrt{1 - (\cos(\cos^{-1} x))^2} \\
 &= \sqrt{1 - x^2}.
 \end{aligned}$$

ACTIVITY 2

Make the indicated trigonometric substitution in the given algebraic expression and simplify it. You can assume $0 < \theta < \frac{\pi}{2}$.

$$(a) \frac{x}{\sqrt{1-x^2}}, \quad x = \sin \theta.$$

$$(b) \frac{1}{x^2\sqrt{4+x^2}}, \quad x = 2 \tan \theta.$$

$$(c) \sqrt{9-x^2}, \quad x = 3 \sin \theta.$$

$$(d) \frac{\sqrt{x^2-25}}{x}, \quad x = 5 \sec \theta.$$

Remark: This is a useful technique in integration.

ACTIVITY 2: SOLUTION

(a) By substituting $x = \sin \theta$, we have

$$\begin{aligned}\frac{x}{\sqrt{1-x^2}} &= \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}} = \frac{\sin \theta}{\sqrt{\cos^2 \theta}} \\ &= \frac{\sin \theta}{\cos \theta} = \tan \theta,\end{aligned}$$

where $\sqrt{\cos^2 \theta} = \cos \theta$ because $\cos \theta > 0$ for $0 < \theta < \frac{\pi}{2}$.

(b) By substituting $x = 2 \tan \theta$, we have

$$\begin{aligned}\frac{1}{x^2 \sqrt{4+x^2}} &= \frac{1}{4 \tan^2 \theta \sqrt{4+4 \tan^2 \theta}} = \frac{1}{8 \tan^2 \theta \sqrt{1+\tan^2 \theta}} \\ &= \frac{1}{8 \tan^2 \theta \sqrt{\sec^2 \theta}} = \frac{1}{8 \tan^2 \theta \sec \theta} \\ &= \frac{\cot^2 \theta \cos \theta}{8},\end{aligned}$$

where $\sqrt{\sec^2 \theta} = \sec \theta$ because $\sec \theta > 0$ for $0 < \theta < \frac{\pi}{2}$.

ACTIVITY 2: SOLUTION

(c) By substituting $x = 3 \sin \theta$, we have

$$\begin{aligned}\sqrt{9 - x^2} &= \sqrt{9 - 9 \sin^2 \theta} = 3\sqrt{1 - \sin^2 \theta} \\ &= 3\sqrt{\cos^2 \theta} = 3 \cos \theta,\end{aligned}$$

where $\sqrt{\cos^2 \theta} = \cos \theta$ because $\cos \theta > 0$ for $0 < \theta < \frac{\pi}{2}$.

(d) By substituting $x = 5 \sec \theta$, we have

$$\begin{aligned}\frac{\sqrt{x^2 - 25}}{x} &= \frac{\sqrt{25 \sec^2 \theta - 25}}{5 \sec \theta} = \frac{5\sqrt{\sec^2 \theta - 1}}{5 \sec \theta} \\ &= \frac{\sqrt{\tan^2 \theta}}{\sec \theta} = \frac{\tan \theta}{\sec \theta} \\ &= \frac{\sin \theta}{\cos \theta} \cos \theta = \sin \theta,\end{aligned}$$

where $\sqrt{\tan^2 \theta} = \tan \theta$ because $\tan \theta > 0$ for $0 < \theta < \frac{\pi}{2}$.

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ADDITION AND SUBTRACTION FORMULAS

ADDITION AND SUBTRACTION FORMULAS

Formulas for sine:

$$\sin(s + t) = \sin s \cos t + \cos s \sin t$$

$$\sin(s - t) = \sin s \cos t - \cos s \sin t$$

Formulas for cosine:

$$\cos(s + t) = \cos s \cos t - \sin s \sin t$$

$$\cos(s - t) = \cos s \cos t + \sin s \sin t$$

Formulas for tangent:

$$\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$$

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

Check out this [video](#) for the proof of the addition formula of sine.

Remarks:

- The formulas for cosine can be derived from those of sine by the identity $\cos x = \sin\left(\frac{\pi}{2} - x\right)$.
- The formulas for tangent are obtained by taking the quotients of the formulas of sine and cosine.

ADDITION AND SUBTRACTION FORMULAS - EXAMPLE

E.g.: The cofunction identities can be derived from the subtraction formulas. For example

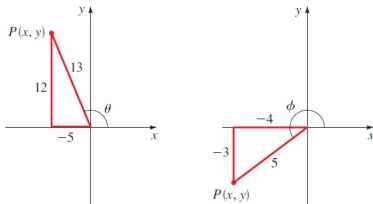
$$\begin{aligned}\cos\left(\frac{\pi}{2} - x\right) &= \cos\left(\frac{\pi}{2}\right)\cos x + \sin\left(\frac{\pi}{2}\right)\sin x \\ &= 0 \cdot \cos x + 1 \cdot \sin x = \sin x.\end{aligned}$$

E.g.: We can compute $\sin(15^\circ)$ by using the subtraction formula:

$$\begin{aligned}\sin(15^\circ) &= \sin(45^\circ - 30^\circ) \\ &= \sin(45^\circ)\cos(30^\circ) - \cos(45^\circ)\sin(30^\circ) \\ &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} \\ &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3} - 1}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}.\end{aligned}$$

ADDITION AND SUBTRACTION FORMULAS - EXAMPLE

E.g.: Suppose θ is in the 2nd quadrant and ϕ is in the 3rd quadrant such that $\sin \theta = \frac{12}{13}$ and $\tan \phi = \frac{3}{4}$, we can evaluate $\sin(\theta + \phi)$ by applying the addition formula of sine. We sketch right triangles to represent the two angles θ, ϕ :



We see that $\cos \theta = -\frac{5}{13}$, $\sin \phi = -\frac{3}{5}$ and $\cos \phi = -\frac{4}{5}$, so

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi = \frac{12}{13} \cdot \left(-\frac{4}{5}\right) + \left(-\frac{5}{13}\right) \cdot \left(-\frac{3}{5}\right) = -\frac{33}{65}.$$

Remark: The addition and subtraction formulas are crucial in proving the derivatives of trigonometric functions.

ACTIVITY 3

Evaluate the following without using a calculator.

(a) $\tan \frac{5\pi}{12}$

(b) $\cos \frac{\pi}{8}$

ACTIVITY 3: SOLUTION

(a)

$$\begin{aligned}
 \tan \frac{5\pi}{12} &= \tan \left(\frac{\pi}{6} + \frac{\pi}{4} \right) \\
 &= \frac{\tan \frac{\pi}{6} + \tan \frac{\pi}{4}}{1 - \tan \frac{\pi}{6} \cdot \tan \frac{\pi}{4}} \\
 &= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}} \cdot 1} \\
 &= \frac{1 + \sqrt{3}}{\sqrt{3} - 1} \\
 &= \frac{(1 + \sqrt{3})^2}{(\sqrt{3} - 1)(\sqrt{3} + 1)} \\
 &= \frac{1 + 2\sqrt{3} + 3}{3 - 1} \\
 &= 2 + \sqrt{3}.
 \end{aligned}$$

ACTIVITY 3: SOLUTION

(b)

$$\cos \frac{\pi}{4} = \cos \left(\frac{\pi}{8} + \frac{\pi}{8} \right)$$

$$\Leftrightarrow \frac{\sqrt{2}}{2} = \cos \frac{\pi}{8} \cos \frac{\pi}{8} - \sin \frac{\pi}{8} \sin \frac{\pi}{8}$$

$$\Leftrightarrow \frac{\sqrt{2}}{2} = \cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8}$$

$$\Leftrightarrow \frac{\sqrt{2}}{2} = \cos^2 \frac{\pi}{8} - \left(1 - \cos^2 \frac{\pi}{8} \right)$$

$$\Leftrightarrow \frac{\sqrt{2}}{2} = 2 \cos^2 \frac{\pi}{8} - 1$$

$$\Leftrightarrow \cos^2 \frac{\pi}{8} = \frac{1}{2} \left(\frac{\sqrt{2}}{2} + 1 \right) = \frac{2 + \sqrt{2}}{4}$$

$$\Leftrightarrow \cos \frac{\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}.$$

ACTIVITY 4

Can you sketch the graph of $f(x) = 3 \sin x - 4 \cos x$? In general, how do we visualize

$$f(x) = a \sin x + b \cos x$$

for $a, b \neq 0$? This kind of functions is the general solution for simple harmonic motions.

(a) We want to find $A \in \mathbb{R}$ and $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$ such that

$$a \sin x + b \cos x = A \sin(x + \alpha).$$

By using the addition formula of sine, find A and α in terms of a, b .

(b) Apply the method in part (a) to $f(x) = 3 \sin x - 4 \cos x$.

(c) Sketch the graph of $f(x) = 3 \sin x - 4 \cos x$. What are the amplitude and period of $f(x)$?

ACTIVITY 4: SOLUTION

(a) By the addition formula of sine,

$$a \sin x + b \cos x = A \sin(x + \alpha) = A \sin x \cos \alpha + A \cos x \sin \alpha.$$

By substituting $x = 0$ and $x = \frac{\pi}{2}$ respectively, we get

$$\Rightarrow \begin{cases} a = A \cos \alpha & (1) \\ b = A \sin \alpha & (2) \end{cases}$$

To solve for $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we divide equation (2) by equation (1) (we can do this because $a \neq 0$):

$$\begin{aligned} \frac{b}{a} &= \frac{A \sin \alpha}{A \cos \alpha} \\ &= \tan \alpha \end{aligned}$$

$$\Rightarrow \alpha = \arctan \left(\frac{b}{a} \right).$$

ACTIVITY 4: SOLUTION

(a) To solve for A , we sum up the square of equations (1) and (2):

$$\begin{aligned} a^2 + b^2 &= A^2 \cos^2 \alpha + A^2 \sin^2 \alpha \\ &= A^2 (\cos^2 \alpha + \sin^2 \alpha) = A^2 \cdot 1 \\ &= A^2. \end{aligned}$$

$$\Rightarrow A = \text{sign}(a) \sqrt{a^2 + b^2}.$$

Special attention is needed to choose the right sign for A such that both $a \sin x + b \cos x$ and $A \sin(x + \alpha)$ have the same sign at $x = 0$ (or any point). Since equation (1) gives $A = \frac{a}{\cos \alpha}$ and $\cos \alpha > 0$ for $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the signs of A and a are equal.

ACTIVITY 4: SOLUTION

(b) Using the results from part (a), for

$$f(x) = 3 \sin x + (-4) \cos x = A \sin(x + \alpha),$$

we have

$$A = \text{sign}(3) \sqrt{3^2 + (-4)^2} = +5,$$

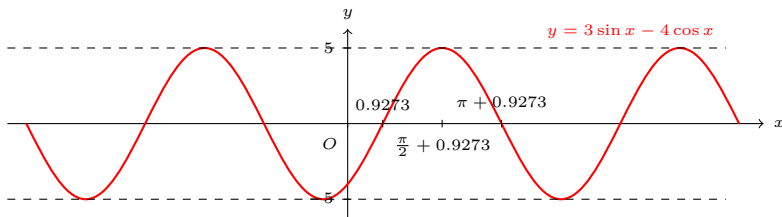
$$\alpha = \arctan\left(\frac{-4}{3}\right) \approx -0.9273 \text{ rad}.$$

Hence

$$f(x) = 3 \sin x - 4 \cos x = 5 \sin(x - 0.9273).$$

ACTIVITY 4: SOLUTION

- (c) Knowing $f(x) = 3 \sin x - 4 \cos x = 5 \sin(x - 0.9273)$, it's easy to sketch the graph of $f(x)$. Just shift the curve $y = \sin x$ to the right by 0.9273 and then stretch vertically by 5.



The amplitude of f is 5 and the period of f is still 2π .

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DOUBLE-ANGLE FORMULAS

The double-angle formulas are just special cases of the addition formulas, with the two angles being the same.

DOUBLE-ANGLE FORMULAS

Formula for sine: $\sin 2x = 2 \sin x \cos x$

Formulas for cosine: $\cos 2x = \cos^2 x - \sin^2 x$
 $= 1 - 2 \sin^2 x$
 $= 2 \cos^2 x - 1$

Formula for tangent: $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Remark: There are 3 versions for the double-angle formula of cosine, because of the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

DOUBLE-ANGLE FORMULAS - EXAMPLE

E.g.: If $\cos x = \frac{4}{5}$ and x is in the 4th quadrant, find $\sin 2x$, $\cos 2x$ and $\tan 2x$. Since x is in the 4th quadrant, $\sin x < 0$. By the Pythagorean identity,

$$\sin^2 x = 1 - \cos^2 x \Rightarrow \sin x = -\sqrt{1 - \left(\frac{4}{5}\right)^2} = -\frac{3}{5}.$$

Hence,

$$\sin 2x = 2 \sin x \cos x = 2 \left(-\frac{3}{5}\right) \frac{4}{5} = -\frac{24}{25},$$

$$\cos 2x = 2 \cos^2 x - 1 = 2 \left(\frac{4}{5}\right)^2 - 1 = \frac{7}{25},$$

$$\tan 2x = \frac{\sin 2x}{\cos 2x} = -\frac{24}{7}.$$

DOUBLE-ANGLE FORMULAS - EXAMPLE

E.g.: We can apply the addition formulas and the double-angle formulas to derive the triple-angle formulas. For instance,

$$\begin{aligned}
 \cos 3x &= \cos(2x + x) \\
 &= \cos 2x \cos x - \sin 2x \sin x \\
 &= (2 \cos^2 x - 1) \cos x - (2 \sin x \cos x) \sin x \\
 &= 2 \cos^3 x - \cos x - 2 \sin^2 x \cos x \\
 &= 2 \cos^3 x - \cos x - 2(1 - \cos^2 x) \cos x \\
 &= 2 \cos^3 x - \cos x - 2 \cos x + 2 \cos^3 x \\
 &= 4 \cos^3 x - 3 \cos x.
 \end{aligned}$$

HALF-ANGLE FORMULAS

By rearranging the double-angle formula of cosine, we can write the squares of $\sin x$ and $\cos x$ in terms of $\cos 2x$. This is very useful in reducing powers in a trigonometric expression and hence simplifying it, in particular, this helps in integration.

$$\sin^2 x = \frac{1 - \cos 2x}{2} \quad \text{and} \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

If we let $u = 2x$, then we get the half-angle formulas

$$\sin^2 \frac{u}{2} = \frac{1 - \cos u}{2} \quad \text{and} \quad \cos^2 \frac{u}{2} = \frac{1 + \cos u}{2}.$$

DOUBLE-ANGLE FORMULAS - EXAMPLE

E.g.: We can express $\sin^2 x \cos^2 x$ in terms of the first power of cosine as follows:

$$\begin{aligned}
 \sin^2 x \cos^2 x &= \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right) \\
 &= \frac{1 - \cos^2 2x}{4} \\
 &= \frac{1 - \frac{1 + \cos 4x}{2}}{4} \\
 &= \frac{1}{8}(1 - \cos 4x).
 \end{aligned}$$

ACTIVITY 5

(a) Evaluate the following without using a calculator.

i) $\cos \frac{\pi}{16}$

ii) $\sin \frac{\pi}{24}$

(b) Express $\tan^2 x$ in terms of $\cos 2x$. Hence, evaluate $\tan \frac{\pi}{16}$.

ACTIVITY 5: SOLUTION

(a) i) We got $\cos \frac{\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$ from activity 3. So

$$\cos^2 \frac{\pi}{16} = \frac{1 + \cos \left(2 \cdot \frac{\pi}{16}\right)}{2}$$

$$= \frac{1 + \cos \frac{\pi}{8}}{2}$$

$$= \frac{1 + \frac{\sqrt{2+\sqrt{2}}}{2}}{2}$$

$$= \frac{2 + \sqrt{2 + \sqrt{2}}}{4}$$

$$\Rightarrow \cos \frac{\pi}{16} = \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2}.$$

Can you deduce the pattern for $\cos \frac{\pi}{2^n}$?

ACTIVITY 5: SOLUTION

(a) ii) We got $\sin \frac{\pi}{12} = \sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}$ from a previous example. So

$$\begin{aligned}\cos \frac{\pi}{12} &= \sqrt{1 - \sin^2 \frac{\pi}{12}} = \sqrt{1 - \left(\frac{\sqrt{6} - \sqrt{2}}{4} \right)^2} \\ &= \frac{1}{4} \sqrt{16 - (6 - 2\sqrt{12} + 2)} = \frac{\sqrt{8 + 4\sqrt{3}}}{4} = \frac{\sqrt{2 + \sqrt{3}}}{2}.\end{aligned}$$

$$\begin{aligned}\sin^2 \frac{\pi}{24} &= \frac{1 - \cos \left(2 \cdot \frac{\pi}{24} \right)}{2} = \frac{1 - \cos \frac{\pi}{12}}{2} \\ &= \frac{1 - \frac{\sqrt{2+\sqrt{3}}}{2}}{2} = \frac{2 - \sqrt{2 + \sqrt{3}}}{4}\end{aligned}$$

$$\Rightarrow \sin \frac{\pi}{24} = \frac{\sqrt{2 - \sqrt{2 + \sqrt{3}}}}{2}.$$

ACTIVITY 5: SOLUTION

(b)

$$\tan^2 x = \frac{\sin^2 x}{\cos^2 x} = \frac{\frac{1-\cos 2x}{2}}{\frac{1+\cos 2x}{2}} = \frac{1-\cos 2x}{1+\cos 2x}.$$

Thus,

$$\begin{aligned} \tan^2 \frac{\pi}{16} &= \frac{1-\cos \left(2 \cdot \frac{\pi}{16}\right)}{1+\cos \left(2 \cdot \frac{\pi}{16}\right)} = \frac{1-\cos \frac{\pi}{8}}{1+\cos \frac{\pi}{8}} \\ &= \frac{1-\frac{\sqrt{2+\sqrt{2}}}{2}}{1+\frac{\sqrt{2+\sqrt{2}}}{2}} = \frac{2-\sqrt{2+\sqrt{2}}}{2+\sqrt{2+\sqrt{2}}} \end{aligned}$$

$$\Rightarrow \tan \frac{\pi}{16} = \sqrt{\frac{2-\sqrt{2+\sqrt{2}}}{2+\sqrt{2+\sqrt{2}}}}.$$

Alternatively, you can also use the result from part (a)i) and the identity

$$\tan^2 \frac{\pi}{16} = \sec^2 \frac{\pi}{16} - 1 = \frac{1}{\cos^2 \frac{\pi}{16}} - 1.$$

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SUMMARY

- Trigonometric Identities
 - fundamental trigonometric identities.
 - simplifying trigonometric expressions.
- Addition and Subtraction Formulas
 - addition and subtraction formulas for sine, cosine and tangent.
- Double-Angle Formulas
 - the double-angle formulas for sine, cosine and tangent.
 - the half-angle formulas.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 7.1, 7.2, 7.3. Sections 6.4, 6.5, 7.4, 7.5 are not covered in this course, but it's good to know.